



MTH 2008 SCIENTIFIC COMPUTING

Danilo Roccatano

SELECTED C++ CODE SOLUTIONS

LAB 1

Objectives of this lab:

1. Learn the basic use of Microsoft Visual Studio 2019

Start to learn about the software life cycle by compiling your first C++ programs

TASK 3: Write your first C++ program

Using the previous examples try to write a C++ program that output the following text on the screen. ***Hint:*** use space between number in the cout to format the numbers in the pyramidal shape.

This is the Pascal's triangle until $n=5$

n

```
0           1
1        1   1
2       1  2  1
3      1 3 3 1
4     1 4 6 4 1
5    1 5 10 10 5 1
```

Made by Danilo.

SOLUTION

NOTE: This is a more elaborate solution of the original task comprising more printed information.

```
/*
 * NAME: LAB 1: TASK 3
 * DESCRIPTION: Write out Pascals Triangle upto n = 5
 * AUTHOR:
 * DATE:
 */
#include <iostream>
using namespace std;

int main() {
    cout << "This is the Pascal\'s triangle upto n=5.    " << endl;
    cout << "The values are obtained by the recursion    " << endl;
    cout << " /  \ \ /  \ \ /  \ \ /  \ \    " << endl;
    cout << " | n |   | n-1 |   | n-1 |       | 0 |    " << endl;
    cout << " |   | = |     | + |     | with |   | =1    " << endl;
    cout << " | k |   | k-1 |   | k   |       | 0 |    " << endl;
    cout << " \ \ /  \ \ /  \ \ /  \ \ /  \ \    " << endl;
    cout << "----+-----k-----" << endl;
    cout << "n   |  0 1 2 3 4 4 5   " << endl;
    cout << "----+-----" << endl;
    cout << endl;
    cout << "0           1           " << endl;
    cout << "1          1 1          " << endl;
    cout << "2         1 2 1         " << endl;
    cout << "3        1 3 3 1        " << endl;
    cout << "4       1 4 6 4 1       " << endl;
    cout << "5      1 5 10 10 5 1     " << endl;
    cout << endl;
    cout << "Made by Danilo Roccatano\a" << endl;
    return 0;
}
```

LAB 2

Objectives of the lab:

- 1. Variables and data types*
- 2. Learn more I/O stream commands: Input from the keyboard*
- 3. Mathematical expressions*

TASK 2.2: Modify the program to add the calculation of the quotient (/) and module (%) operation (remainder) of the two numbers.

SOLUTION

```
/* NAME: Lab2.2.2.cpp
 * DESCRIPTION: use the I/O stream from the keyboard and math operators
 * AUTHOR:
 * DATE:
 */

#include <iostream>
using namespace std;

int main() {

    int num1, num2; // declare variables
    int out;

    cout << "Enter two integers: "; // prompt user

    cin >> num1 >> num2;           // read the two values from keyboard

    // output the results
    out = num1 + num2;
    cout << "\nThe sum is          :" << out;
    out = num1 * num2;
    cout << "\nThe product is       :" << out;
    out = num1 - num2;
    cout << "\nThe difference is  :" << out;
    out = num1 / num2;
    cout << "\nThe quotient is    :" << out;
    out = num1 % num2;
    cout << "\nThe reminder is     :" << out;
    cout << "\nNote that the module correspond to the following operation:" <<
endl;
```

```

out = num1-num2*(num1/num2);
cout << "out = num1-num2*(num1/num2) = " << out << endl;
return 0;
}

```

TASK 2.2: Modify the program to implement the following calculations.

1. **Calculate Remainder:** Prompt the user for two integers as input and use the modulo operator to calculate the remainder when the first number is divided by the second. Display the remainder using **cout**.

```

/* NAME: Lab3.2.1.cpp
 * DESCRIPTION: Calculate the remainder when one integer is
 * divided by another.
 * AUTHOR: Danilo Roccatano
 * DATE:
 */

#include <iostream>

using namespace std;

int main() {
    int num1, num2;

    // Prompt the user for input
    cout << "Enter the first integer: ";
    cin >> num1;

    cout << "Enter the second integer: ";
    cin >> num2;

    // Calculate and display the remainder
    int remainder = num1 % num2;
    cout << "The remainder of " << num1 << " divided by " << num2 << " is
" << remainder << endl;

    return 0;
}

```

2. **Calculate Gravitational Force:** Create a program that prompts the user for the masses of two objects and the distance between them, and then calculate the gravitational force between them using the formula **Force = (G * mass1 * mass2) / (distance*distance)**, where **G**= 6.67430×10⁻¹¹ N·m²·kg⁻² is the gravitational constant. Display the result using **cout**.

```

/* NAME: Lab2.2.4.cpp
 * DESCRIPTION: Calculate the gravitational force between two objects.
 * AUTHOR: Danilo Roccatano
 * DATE: Sept. 2023
 */

```

```
#include <iostream>

using namespace std;

int main() {
    double mass1, mass2, distance;
    const double G = 6.67430e-11;

    // Prompt the user for input
    cout << "Enter the mass of the first object (in kg): ";
    cin >> mass1;

    cout << "Enter the mass of the second object (in kg): ";
    cin >> mass2;

    cout << "Enter the distance between the objects (in meters): ";
    cin >> distance;

    // Calculate gravitational force
    double force = (G * mass1 * mass2) / (distance * distance);

    // Display the result
    cout << "The gravitational force is: " << force << " N" << endl;

    return 0;
}
```



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LAB 3

Objectives of the lab:

1. *Overflow and Underflow of integers*
2. *Mathematical function and expressions in C++*

1. Mathematics function in C++

Task 2.1: Use C++ as a calculator: Test simple mathematical expressions

The following program for each function of the Table I print its scope and the results of the example given in the table.

```

#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    cout << "The function sqrt(n) calculate the square root of n." << endl;
    cout << "Ex. sqrt(7.0)      =" <<sqrt(7.0) << endl;
    cout << "Ex. (7+sqrt(7.0))/3.0  =" <<(7+sqrt(7.0))/3.0 << endl;
    cout << "The function log(n) calculate the natural logarithm of n." <<
endl;
    cout << "Ex. log(1.1)          =" <<log(1.1) << endl;
    cout << "The function log10(n) calculate the logarithm in base 10 of n."
<< endl;
    cout << "Ex. log10(1.1)         =" <<log(1.0) << endl;
    cout << "The function pow(m,n) calculate the power n of m." << endl;
    cout << "Ex. pow(2.5,3.0)      =" <<pow(2.5,3.0) << endl;
    cout << "The function tanh(n) calculate the hyperbolic tangent of n." <<
endl;
    cout << "Ex. tanh(0.342)      =" <<tanh(0.342) << endl;
    cout << "The function fmod(m,n) calculate the remainder of m divided n." <<
endl;
    cout << "Ex. fmod(11.0,4.0)    =" <<tanh(0.342) << endl;
    cout << "The function abs(n) calculate the absolute value of n." << endl;
    cout << "Ex. abs(-9)          =" << abs(-9) << endl;
    cout << "The function fabs(n) calculate the absolute value of a floating
point number n." << endl;
    cout << "Ex. fabs(-4.5)       =" << fabs(-4.5) << endl;
    cout << "The function ceil(n) calculate the larger whole number that
approximate n." << endl;
    cout << "Ex. ceil(4.1)        =" <<ceil(4.1) << endl;
    cout << "The function floor(n) calculate the smaller whole number that
approximate n." << endl;
    cout << "Ex. floor(8.8)       =" <<floor(8.8) << endl;
    cout << "The function cbrt(n) calculate the cube root of n." << endl;
    cout << "Ex. cbrt(10.0)      =" <<cbrt(10.0) << endl;

}

```

Common C++ Functions

Function	Header File	Purpose	Parameter(s) Type	Result
<code>abs(x)</code>	<code><cstdlib></code>	Returns the absolute value of its argument: <code>abs(-7) = 7</code>	<code>int</code>	<code>int</code>
<code>ceil(x)</code>	<code><cmath></code>	Returns the smallest whole number that is not less than <code>x</code> : <code>ceil(56.34) = 57.0</code>	<code>double</code>	<code>double</code>
<code>cos(x)</code>	<code><cmath></code>	Returns the cosine of angle <code>x</code> : <code>cos(0.0) = 1.0</code>	<code>double</code> (radians)	<code>double</code>
<code>exp(x)</code>	<code><cmath></code>	Returns e^x , where $e = 2.718$: <code>exp(1.0) = 2.71828</code>	<code>double</code>	<code>double</code>
<code>fabs(x)</code>	<code><cmath></code>	Returns the absolute value of its argument: <code>fabs(-5.67) = 5.67</code>	<code>double</code>	<code>double</code>

Common C++ Functions

Function	Header File	Purpose	Parameter(s) Type	Result
<code>floor(x)</code>	<code><cmath></code>	Returns the largest whole number that is not greater than <code>x</code> : <code>floor(45.67) = 45.00</code>	<code>double</code>	<code>double</code>
<code>pow(x, y)</code>	<code><cmath></code>	Returns <code>x^y</code> ; If <code>x</code> is negative, <code>y</code> must be a whole number: <code>pow(0.16, 0.5) = 0.4</code>	<code>double</code>	<code>double</code>
<code>tolower(x)</code>	<code><cctype></code>	Returns the lowercase value of <code>x</code> if <code>x</code> is uppercase; otherwise, returns <code>x</code>	<code>int</code>	<code>int</code>
<code>toupper(x)</code>	<code><cctype></code>	Returns the uppercase value of <code>x</code> if <code>x</code> is lowercase; otherwise, returns <code>x</code>	<code>int</code>	<code>int</code>

Source: http://www.cs.fsu.edu/~cop3014p/lectures/ch6_new/index.html

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TASK 2.2: Use C++ as a calculator: evaluate complex expressions

```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    cout << "Expression 1:" << 2./15.-(1./3.-sin(0.5))/(-5./13)+(1./3.-
    tanh(0.3))/(-1./7.-2.0) << endl;
    cout << "Expression 2:" << fabs(pow((-9.0*pow(-5.0,-2.0)-
    1.0),sin(1.0)))*(pow(4.0/2.0,2.0)+pow(1.0/3.0,-2.0)-cbrt(-5))/sqrt(17.0/3.0)
    << endl;
}
```

Note that the expression 2 give as results NaN (not a number) as the

$$(-9(-5)^{-2} - 1)^{\sin(1.0)}$$

is not defined as a real number.

TASK 2.3: Use C++ as a calculator: evaluate complex expressions containing variables

```
#include <iostream>
#include <cmath> // Required for hyperbolic functions
using namespace std;

int main()
{
    double x, y;
    int select;

    // Display the list of functions
    cout << "FUNCTION LIST" << endl;
    cout << "1) csch(x) = 1 / sinh(x)" << endl;
    cout << "2) sech(x) = 1 / cosh(x)" << endl;

    // Prompt the user to select a function
    cout << "Select a number (1 or 2): ";
    cin >> select;

    // Request the value of x
    cout << "Enter a value for x: ";
    cin >> x;
    cout << "For the value of x = " << x << endl;
    cout << "The function ";

    // Calculate the chosen function based on the user's selection
    if (select == 1) {
        y = 1 / sinh(x); // csch(x)
        cout << "y = 1 / sinh(x)";
    }
    else if (select == 2) {
        y = 1 / cosh(x); // sech(x)
        cout << "y = 1 / cosh(x)";
    }
    else {
        cout << "Invalid selection. Please run the program again." << endl;
        return 1; // Exit if the selection is invalid
    }
}
```

```
// Output the result
cout << " is equal to " << y << endl;
return 0;
}
```



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LAB 4

Objectives of the lab:

1. *Mathematical expressions in C++*
2. *Control structure if-else*

Use of mathematical expression in C++ and the conditional statements *if then else*

TASK 1: evaluate a piecewise defined function

- Write in the logbook the mathematical expression of the function calculated by the program using the mathematical notation.

$$y = \begin{cases} 1 - e^x & 0.5 < x < 2 \\ x^3 + \sqrt{x} & x \geq 2 \\ |x + 5| & x \leq 0.5 \end{cases}$$

- Modify the program to calculate the following piece-wise defined function:

$$y = \begin{cases} -1 - \frac{\pi}{2} & x < -\frac{\pi}{2} \\ \sin(x) + x & -\frac{\pi}{2} \leq x \leq 0 \\ x & x > 0 \end{cases}$$

```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x,y;
    cout << " Enter the value of x: ";
    cin >> x;

    if (x < -M_PI/2. ) {
        cout << "The value of x < -pi/2 and\n";
        cout << "y := -1-pi/2= ";
        y= -1 -M_PI/2.0;
    }
    else if ( x > -M_PI/2.0 && x <=0.0 ) {
        cout << "The value of -pi/2 < x <= 0 and \n";
        cout << "y := sin(x)+x = ";
        y= sin(x)+x;
    } else {
        cout << "The value of x > 0 and \n";
        cout << "y := x = ";
        y= x ;
    }
    cout << y << endl;
    return 0;
}
```

TASK 2: Check input values

In this Task, you are going to learn how to check the input data from the *cin* stream.

Modify the code written in the LAB 3 for TASK 2.3 (use C++ as a calculator: evaluate complex expressions containing variables) to check of the user input for the expression 3 is equal to zero.

```

#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    const double val=0.0;
    const double smallnum=10e-12;
    double x,y;

    /*
    * Calculation of the hyperbolic arccosecant function
    */

    cout << "Calculation of the hyperbolic arccosecant function. \n";

    cout << " Enter any value of x but different from x=0.0: ";
    cin >> x;

    if (fabs(x-val) < smallnum) {
        cout << "ERROR: value not allowed!\n Please Try again ... \n";
        return 1;
    }
    cout << "For the value of  x = " << x << endl;

    y= log(1.0/x +sqrt(x*x+1)/fabs(x));

    cout << "acsch(" << x << ") = " << y << endl;
    return 0;
}

```

TASK 3: use C++ as a calculator

Modify the given code to select the function to calculate. A C++ code template is given below.
Add to the list such also the function:

$$y = \frac{1}{\cosh(x)}$$

```

#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x, y;
    int select;

```

```

// Display the list of functions
cout << "FUNCTION LIST" << endl;
cout << "1) csch(x) = 1 / sinh(x)" << endl;
cout << "2) sech(x) = 1 / cosh(x)" << endl;

// Prompt the user to select a function
cout << "Select a number (1 or 2): ";
cin >> select;

// Request the value of x
cout << "Enter a value for x: ";
cin >> x;
cout << "For the value of x = " << x << endl;
cout << "The function ";

// Calculate the chosen function based on the user's selection
if (select == 1) {
    y = 1 / sinh(x); // csch(x)
    cout << "y = 1 / sinh(x)";
}
else if (select == 2) {
    y = 1 / cosh(x); // sech(x)
    cout << "y = 1 / cosh(x)";
}
else {
    cout << "Invalid selection. Please run the program again." << endl;
    return 1; // Exit if the selection is invalid
}

// Output the result
cout << " is equal to " << y << endl;
return 0;
}

```



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LAB 5

Objectives of the lab:

Experimenting with control statements:

- *The Switch statement.*
- *The while loop.*
- *The do-while loop.*

EXERCISE 1

The Switch control statement

Task 1.1: take confidence with the syntax of the Switch statement by copying and running the following program.

```
#include <iostream>
using namespace std;

int main()
{
    unsigned int i = 3;
    unsigned int n;

    switch ( i )
    {
        case 0:n=1; break; // executed for i==0 and i==1
        case 1:n=1; break;
        case 2:n=2; break;
```

```

        case 3:n=6; break;
        case 4:n=24; break;
        default:n=0; break;
    }
    cout<< "n= "<< n <<endl;
    return 0
}

```

Task 1.2: *Control the function calculator developed in Task 3 of LAB 4 using the switch statement. This is the partial C++ code function calculator in which the if-else control has been substituted with the switch control statement.*

1. Copy and run the program.
2. Modify by adding the missing functions.

```

#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    const double val=2.0;
    const double smallnum=10e-12;
    double x,y;
    int select;

    cout << "FUNCTION LIST" << endl;
    cout << "1) asinh(x)" << endl;
    cout << "Select a number" << endl;
    cin >> select;

    if (select >0 && select < 4) {
        cout << " Enter a value of x: ";
        cin >> x;
        cout << "For the value of  x = " << x << endl;
    }

    /*
     * Starting of the switch selection
     */

    switch (select)
    {
        case 1:
        {
            y= 1/sinh(x);
            cout << "The function y= 1/sinh(x)";
        }
        break;
        case 2:
        {
< ADD HERE THE CODE FOR THE 2nd FUNCTION >

        }
    }
}

```

< ADD HERE THE CODE FOR THE 3rd FUNCTION >

```
        default: {
            cout << "ERROR: the selection is not available" << endl;
            return 1;
        }
    }
    cout << "is equal to ";
    cout << y << endl;
    return 0;
}
```

EXERCISE 2

Task 2.1: *The while-loop control statement.*

Copy the following program to calculate some statistical properties from the set of numbers.

```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    int count = 0;
    double x;
    double sum = 0;
    double min=1e10; // set the variable min to a large number
    double mean;
    // ask the user to input 6 number from the keyboard
    setprecision(10);
    cout << "Enter 6 numbers separated by a single space, press <Enter>: ";
    while( count < 6)
    {
        cin >> x;
        sum = sum + x;
        // check if x is small than min and in case assign min to x
        if (x<min) min=x;
        count++;
    }
    cout << endl;
    average = sum/6;
    cout << "The minimum number is " << min << endl;
    cout << "The mean value is " << mean << endl;
    return 0;
}
```

Modify the program to add the following calculations:

- *Find the largest of the input numbers.*

- *Calculate standard deviation using the Naïve algorithm¹.*

HINT: Variance is given by

$$s^2 = \left(\frac{\sum_1^n x_i^2}{n} - \left(\frac{\sum_1^n x_i}{n} \right)^2 \right) \frac{n}{n-1}$$

with n the total number of data and x_i , i^{th} datum. To implement the algorithm in the given code, you must also calculate the sum of the square of each input data value. Once all the data has been inserted the program will use the given formula to evaluate the variance (s^2). The standard deviation, which is the square root of the variance.

Task 2.2: *The do-while control statement.*

Modify the following program to get the number input from the user.

```
#include <iostream>
using namespace std;

int main() {
    int n=1;
    int i=1;

    do
    {
        n *= i;
        ++i;
    } while (i < 10);
    cout << "The factorial of 10 is " << n << endl;
}
```

Task 2.3: calculation of the factorial using the do-while loop.

- Copy and run the following program to calculate the factorial of 10.
- Modify the following program to input by the keyboard to calculate the factorial.
- Check the biggest number that you can input before getting the wrong results.
- Think about the possible cause of the error and where do you need to modify the program to get the correct factorial with a larger integer.
- Draw the flowchart of the modified program.

¹ See https://en.wikipedia.org/wiki/Algorithms_for_calculating_variance

```

#include <iostream>
using namespace std;

int main() {
    int n=1;
    int i=1;

    do
    {
        n *= i;
        ++i;
    } while ( i < 10 );
    cout << "The factorial of 10 is " << n << endl;
}

```

EXERCISE 3

Task 3.1: *Implementation of the Babylonian algorithm for calculating square roots.*

The Babylonian algorithm is a method of finding square roots. Let's have a go at coding up the Babylonian method (see details here:

https://en.wikipedia.org/wiki/Methods_of_computing_square_roots#Babylonian_method)

Let x and a be positive integers. The sequence

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$$

should converge to the square root of a .

The program *needs to*

- a) *input the number a from the keyboard.*
- b) *define an initial for x_1 (for example $x_1 = 1$).*
- c) *Use a while or do-while loop that generates new values of x_n .*
- d) *Define a stopping criterion. Hint: it is convenient to take the difference (or square of the difference) of the new x_{n+1} and the previous value x_n .*

Coursework Assignment Questions (Set 1/4)

Due date: November 11, 2024

Please submit your answers to all questions as a single file for Coursework 1. The C++ program code and the report (in MS Word or PDF format) should be uploaded to Blackboard. If the program source code is not submitted either as a separate file or included within the report, the exercise will be marked as 'not attempted' (i.e., zero marks), even if numerical results and discussions are provided. All source code must be copied and pasted into the document—screenshots will not be accepted.

Set 1: Two questions for a total of 20 marks

Q1 (Marks 10): Modify the program given in the Task 2.1 of the Lab 2 to implement the following calculations.

1. **Calculate the Area and Circumference of a Circle:** Prompt the user for the radius of a circle to two variables and calculate its area and circumference (or perimeter). Display the results using `cout`.
2. **Calculate Simple Interest:** Create a program that calculates simple interest based on the principal amount, interest rate, and time (in years) entered by the user. Use the formula $\text{Simple Interest} = (\text{Principal} * \text{Rate} * \text{Time}) / 100$. Display the calculated interest using `cout`.

Q2 (Marks 10): Mixed Questions on Computing and Programming Basics

1. **True or False:** In a computer, the main memory stores numbers and letters.
2. The printer is an output _____.
3. A C++ source program must be saved in a text file with a(n) _____ extension.
4. What is the role of a compiler?
5. A byte consists of how many bits?
6. FACADE_{16} is a number expressed in the _____ base.
7. What is the decimal equivalent of FACADE_{16} ?
8. **True or False:** The keyword `int` in C++ is used to declare a variable of type integer.
9. What is wrong with this C++ program?

```
#include <iostream>
int main()
{
    // prints the name of two famous scientists
    cout << "1) Alan Turing \n 2) John von Neumann\n"
}
```

10. Identify and fix the error in the following code.

```
#include <iostream>

using namespace std;

int main() {
    float num1, num2;
    cout << "Enter two numbers: ";
    cin >> num1 num2;
    cout << "The sum is: " << num1 + num2;
    cout << "\nThe product is: " << num1 * num2;
    cout << "\nThe difference is: " << num1 - num2;
    return 0;
}
```

Coursework Assignment Questions (Set 2/4)

Due date: November 11, 2024

Please submit your answers to all questions as a single file for Coursework 1. The C++ program code and the report (in MS Word or PDF format) should be uploaded to Blackboard. If the program source code is not submitted either as a separate file or included within the report, the exercise will be marked as 'not attempted' (i.e., zero marks), even if numerical results and discussions are provided. All source code must be copied and pasted into the document—screenshots will not be accepted.

Set 2: Two questions for a total of 30 marks

Q3 (Marks 20): Use C++ as a calculator to Evaluate Inverse Hyperbolic Functions

Write a C++ program that calculates the following two mathematical functions:

1. Inverse Hyperbolic Tangent ($\operatorname{atanh}(x)$):

$$\operatorname{atanh}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

Domain: $-1 < x < 1$

2. Inverse Hyperbolic Cosine ($\operatorname{acosh}(x)$)

$$\operatorname{acosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right)$$

Domain: $x \geq 1$

Program Requirements:

- The program must request the user to input the value of x using the *cin* command.
- Display a message informing the user of the valid domain for each function.

Example Output:

This program calculates the inverse hyperbolic tangent (atanh) and inverse hyperbolic cosine (acosh).

Note: The valid domain for $\operatorname{atanh}(x)$ is $-1 < x < 1$.

Enter a value for x : 0.5

$\operatorname{atanh}(0.5) = 0.549306$

Note: The valid domain for $\operatorname{acosh}(x)$ is $x \geq 1$.

Enter a value for x : 0.5

$\operatorname{acosh}(0.5)$ is not defined for $x < 1$.

Submission Instructions:

- Write your code in a .cpp file.
- Provide explanations for each task as comments in your code.
- Ensure your code compiles without errors.
- Avoid using AI tools for code generation. Your submission will be checked for originality, and any signs of AI usage will be flagged.

Total: 20 marks for the programming exercise

1. **User Input (5 marks):**
 - Correctly prompts for and reads the value of x using `cin`: **5 marks**.
 - Incorrectly prompts or does not read input: **0 marks**.
2. **Domain Validation (5 marks):**
 - Correctly displays valid domain messages for both functions: **5 marks**.
 - Missing or incorrect domain validation messages: **0 marks**.
3. **Implementation of $\operatorname{atanh}(x)$ (5 marks):**
 - Correctly implements and calculates the inverse hyperbolic tangent function: **5 marks**.
 - Incorrect implementation or calculation error: **0 marks**.
4. **Implementation of $\operatorname{acosh}(x)$ (5 marks):**
 - Correctly implements and calculates the inverse hyperbolic cosine function: **5 marks**.
 - Incorrect implementation or calculation error: **0 marks**.

Q4 (Marks 10): Mixed Questions on Computing and Programming Basics

Multiple choice (2 mark each)

- 1) Which of the following is the primary function of an Operating System?
 - a) A program that helps organize computer files.
 - b) An interface that allows interaction between hardware and the user.
 - c) Software responsible for compiling source code.
 - d) A set of utilities for system backup.
- 2) When developing software, what should be your first step?
 - a) Writing the source code.
 - b) Designing and planning the solution.
 - c) Testing the compiled program.
 - d) Optimizing the object code.
- 3) What does a compiler do in the software development process?
 - a) Translates a high-level program into machine language.
 - b) Organizes data in a specific way.
 - c) Runs the source code line by line.
 - d) Compiles all the object programs into one file.

True/False (1 mark each)

- 1) A computer algorithm is a step-by-step process to solve a problem. (T/F)
- 2) C++ requires every variable to be declared before it is used. (T/F)
- 3) The main function in a C++ program must always return an integer. (T/F)
- 4) In C++, comments can be written using both `//` and `/* */` styles. (T/F)

Coursework Assignment Questions (Set 3/4)

Due date: November 11, 2024

Please submit your answers to all questions as a single file for Coursework 1. The C++ program code and the report (in MS Word or PDF format) should be uploaded to Blackboard. If the program source code is not submitted either as a separate file or included within the report, the exercise will be marked as 'not attempted' (i.e., zero marks), even if numerical results and discussions are provided. All source code must be copied and pasted into the document—screenshots will not be accepted.

Set 3: Two questions for a total of 25 marks

Q5 (Marks 25): program a quadratic equation solver.

Write a C++ program to find the two solutions of a quadratic equation of the form:

$$ax^2 + bx + c = 0$$

using the quadratic formula:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hint:

You need to test the discriminant ($\Delta = b^2 - 4ac$) to handle different cases:

- If $\Delta > 0$, there are two distinct real solutions.
- If $\Delta = 0$, there is one real solution (a double root).
- If $\Delta < 0$, the solutions are complex (not real).

Here's a code snippet to help guide you, where the three dots (...) indicate missing parts to be filled in the final implementation.

```

#include <iostream>
#include <cmath>
using namespace std;
. . .

d = b*b - 4*a*c; \\ The discriminant

// testing the discriminant for real solutions to a quadratic

if (d >= 0.0) {
    // real solutions
    root1 = (-b + sqrt(d)) / (2.0*a);
    . . .
}
. . .

```

Submission Instructions:

- Write your code in a .cpp file.
- Provide explanations for each task as comments in your code.
- Ensure your code compiles without errors.
- Avoid using AI tools for code generation. Your submission will be checked for originality, and any signs of AI usage will be flagged.

Total: 25 marks for the programming exercise

Functionality (15 marks)

1. Non-Functional Code:
Some evidence of programming, but the code does not compile or run: max 3 marks.
2. Functional Code with Errors:
The program compiles and runs, but the results are incorrect or inconsistent: max 3 marks.
3. Partially Correct Functionality:
The program works, and the results are partially correct or complete: max 3 marks.
4. Fully Correct Functionality:
The program works correctly, and the results are accurate and complete: max 3 marks.
5. Exceptional Functionality:
The program is exceptionally well done, featuring extensions and/or additional functionalities: max 3 marks.

Code Quality (10 marks)

Well-Described and Commented Code:

The code is well-structured, with clear, concise comments explaining the logic and flow of the program.

Coursework Assignment Questions (Set 4/4)

Due date: November 11, 2024

Please submit your answers to all questions as a single file for Coursework 1. The C++ program code and the report (in MS Word or PDF format) should be uploaded to Blackboard. If the program source code is not submitted either as a separate file or included within the report, the exercise will be marked as 'not attempted' (i.e., zero marks), even if numerical results and discussions are provided. All source code must be copied and pasted into the document—screenshots will not be accepted.

Set 3: One question of 25 marks

Q6 (Marks 25): C++ Variable Manipulation and Output.

Write C++ statements that accomplish the following tasks. Ensure that you understand each task and can explain your code.

Tasks:

1) Declare and Initialize Variables

- Declare two integer variables, `x` and `y`. Initialize `x` to 25 and `y` to 18.
- Declare and initialize a double variable named `payRate` to 12.50.
- Create an integer variable called `firstNum` and initialize it to a value of your choice. Then, declare another integer variable `tempNum` and copy the value of `firstNum` into `tempNum`.

2) Swap Values

Write the code to swap the contents of the integer variables `x` and `y`. You may declare additional variables as necessary. Explain how your swap works in a comment.

3) Output Expressions

Assume `x` and `y` are now double variables. Write a C++ statement to output the contents of `x`, `y`, and the result of the expression $x + 12 / y - 18 + x^3$.

Explain the output in a comment above the line of code.

Submission Instructions:

- Write your code in a `.cpp` file.
- Provide explanations for each task as comments in your code.
- Ensure your code compiles without errors.
- Avoid using AI tools for code generation. Your submission will be checked for originality, and any signs of AI usage will be flagged.

Total: 25 marks for the programming exercise

Functionality (15 marks)

1. Non-Functional Code:
Some evidence of programming, but the code does not compile or run: max 3 marks.
2. Functional Code with Errors:
The program compiles and runs, but the results are incorrect or inconsistent: max 3 marks.
3. Partially Correct Functionality:
The program works, and the results are partially correct or complete: max 3 marks.
4. Fully Correct Functionality:
The program works correctly, and the results are accurate and complete: max 3 marks.
5. Exceptional Functionality:
The program is exceptionally well done, featuring extensions and/or additional functionalities:
max 3 marks.

Code Quality (10 marks)

Well-Described and Commented Code:

The code is well-structured, with clear, concise comments explaining the logic and flow of the program.